1. Evidence review report: EXP-CMFB-cea1d2 (DEC-20261019-f33f3e)

Decision report for the Coordinator's evidence review. Source written without a shell; the calling session renders `fig_review.svg`, `fig_review.png` and this PDF from `plot.py` and `graph_data.json`. Numbers are read from `RUN-CMFB-2010d3/raw-result.json` by key, from the two review reports, or are arithmetic on those values (marked *derived*). No number here was produced by running code in this review.

1.1. Decision, strength, tier

Decision	**refine** (DEC-20261019-f33f3e). Not official until the ledger archive commit is verified by the dispatcher.
Evidence	EV-CMFB-e3d1ec: direction **supports** H-CMFB-9b2eed (a null prediction), strength **preliminary**, claim tier **toy**, proof status `empirical_only` (the 1.582 coverage ceiling is a separate derivation, KN-FIND-007).
Hypothesis	H-CMFB-9b2eed: specified to **analyzed** (not supported).
Experiment, runs	EXP-CMFB-cea1d2; RUN-CMFB-2373be (t1), RUN-CMFB-5905c0 (t2), RUN-CMFB-66b0fe (t3), RUN-CMFB-3dc120 (t4, C16 and C20), RUN-CMFB-42a578 (t5, C24 FAM), RUN-CMFB-e13f0e (t6, C24 N-A), RUN-CMFB-c71d35 (t7, C24 N-B), RUN-CMFB-2010d3 (t8, analysis).
Reviews	VAL-20261019-1b97f5 (task TASK-20261019-bbb351, verdict passed); RT-20261019-520123 (verdict conclusion_requires_narrowing). Review plan, written after both verdicts: `experiments/EXP-CMFB-cea1d2/reviews/review_plan.yaml`.
Question	RQ-CMFB-33610e (active; no direction closed).

1.2. Exact scoped claim

No coverage difference between the small-x liftable-window factor base and a matched random base exceeding the predefined resolution delta = 0.5 percent was observed over the tested instances (107 prime-order curves with 2^14 < p <= 2^25: 27 FAM members of p = 5t^2 + 19t + 19, 40 other D = −19 crater-type curves, 40 random curves), base (the first ceil((6N)^(1/3)) abscissae with f(x) a nonzero square, one canonical-sign point each, integral Weierstrass chart), counting convention (m = 3, unsigned multisets, targets uniform over Z_N, exact counting, 200 matched random bases per member) and resource budget (16 minutes wall, one pass, seeds 7001 to 7010). At C24 (2^22 < p <= 2^25; FAM 13 of 13 members in the band) every recorded 95 percent interval of the mean coverage ratio minus 1 lies inside +-0.04 percent; at C20 inside [−0.074, +0.057] percent under the contract-text powered set. C16 (FAM n = 4) is underpowered and enters no conclusion.

This is a statement at resolution 0.5 percent for the stated arms, cells, chart, base and convention, not a statement that no effect exists, and not that no factor base can beat a random one (a Sidon-type base reaches +10.7 percent, EV-FBG-001). The coverage ceiling (ratio at most 1.582) is the theorem KN-FIND-007, not a result of this run. Untested: the CM structure (endomorphism chain, isogeny pullback, canonical lift, other models, joint windows, the floor curve), the signed x-only convention, growth in N, decomposition cost, relation rank, linear algebra. No ECDLP speed, rho comparison or pair-46-scale claim; no cross-method ranking is made, so section 5a pricing does not apply.

1.3. Graph

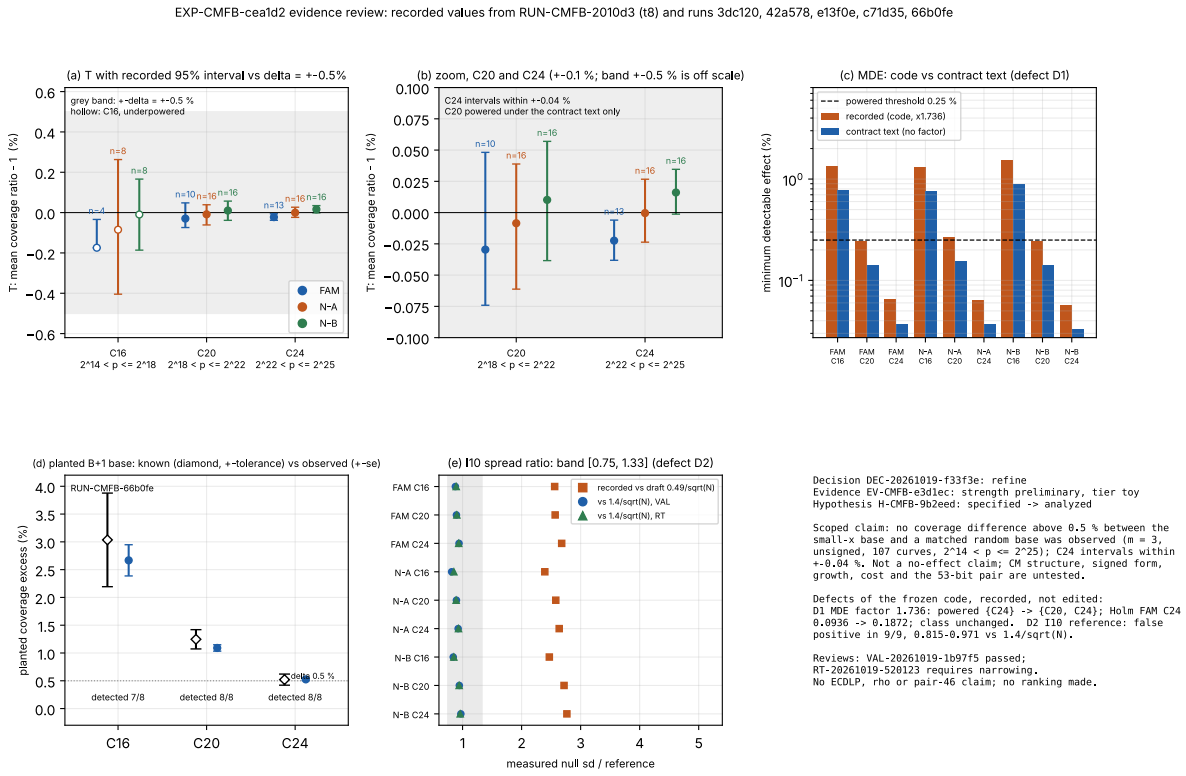


Figure 1: (a) recorded T with 95 percent interval per arm and cell against the grey band delta = +-0.5 percent; hollow markers are C16 (underpowered under every reading). (b) zoom on C20 and C24. (c) minimum detectable effect as recorded by the code (factor sqrt(3) x 1.0025 = 1.736) against the contract text, with the 0.25 percent powered threshold (defect D1). (d) planted B+1 base: known excess with the gate tolerance against the observed pooled excess with its standard error, detections of 8 curves (RUN-CMFB-66b0fe). (e) I10 null-spread ratio: recorded against the superseded 0.49/sqrt(N) reference, and against the frozen 1.4/sqrt(N) as recomputed by the validator and by the red team (defect D2). Sources: RUN-CMFB-2010d3 raw-result.json (tables, cells, sensitivity), VAL-20261019-1b97f5, RT-20261019-520123; data in `graph_data.json`; script `plot.py`.

1.4. Observation

The run is valid and complete (8 of 8 trials; 72 of 72 artifact hashes recomputed; all 26,331 coverage counts recounted by independent code with 0 mismatches; all eight trials rebuilt from scratch with identical raw results; VAL M4, M5, M9). Recorded T in percent, with sample size n:

Arm	Cell	n	T (%)	95 % interval (%)	rand. p	MDE code / contract (%)	Holm p recorded / contract text
FAM	C16	4	−0.174	[−0.174, −0.034]	0.411	1.332 / 0.767	none / none
FAM	C20	10	−0.030	[−0.074, +0.048]	0.441	0.243 / 0.140	none / 1.0
FAM	C24	13	−0.022	[−0.038, −0.006]	0.031	0.065 / 0.037	0.0936 / 0.1872
N-A	C16	8	−0.085	[−0.405, +0.263]	0.686	1.318 / 0.759	none / none
N-A	C20	16	−0.009	[−0.061, +0.039]	0.843	0.268 / 0.154	none / 1.0
N-A	C24	16	−0.000	[−0.024, +0.027]	0.969	0.063 / 0.037	0.9691 / 1.0
N-B	C16	8	−0.009	[−0.186, +0.166]	0.969	1.537 / 0.885	none / none
N-B	C20	16	+0.010	[−0.038, +0.057]	0.795	0.245 / 0.141	none / 1.0
N-B	C24	16	+0.016	[−0.001, +0.035]	0.079	0.057 / 0.033	0.1589 / 0.3972

Contract-text Holm values are *derived* from the recorded randomization p with six tests (C20 and C24 powered): 6 x 0.0312 = 0.1872, 5 x 0.0794 = 0.3972, all others 1.0 (equal to VAL M7). The contract-text MDE column is VAL D1 and RT-O2, equal to recorded divided by 1.7364. Run IDs by row: C16 and C20 rows RUN-CMFB-3dc120; C24 FAM RUN-CMFB-42a578, N-A RUN-CMFB-e13f0e, N-B RUN-CMFB-c71d35; analysis RUN-CMFB-2010d3.

Gates: planted base detected 7 of 8 (C16), 8 of 8 (C20), 8 of 8 (C24), pooled ratio 1.02667 vs 1.03035, 1.01091 vs 1.01247, 1.00525 vs 1.00521, all within tolerance; qr_window R_pool 1.9940 to 2.0009 and R_wrong 0.9970 to 1.0006 in all nine pools; M0 holds on every base. Recorded classification NULL_AT_RESOLUTION, no Holm rejection, three C24 contrasts include 0, growth not assessable.

1.5. Comparison

Every interval is inside the predefined +-0.5 percent band (decision rule R3), and in the cells the contract text calls powered inside +-0.08 percent. The instrument could have failed: the planted base was recovered at +0.525 percent against a known +0.521 percent at C24. The classification is the same under the recorded powered set {C24} and the contract-text set {C20, C24}. The nine null-spread ratios against 1.4/sqrt(N) are 0.815 to 0.971, inside the accepted band [0.75, 1.33]; the recorded 2.39 to 2.77 compare against a superseded reference. The smallest of nine recorded randomization p (0.031, FAM C24, a deficit) has probability about 0.25 as a minimum under the null (*derived*: 1 - 0.969^9). Against the preregistered prediction T = 0 the data agree; against the preregistered success criterion (outcome NULL_AT_RESOLUTION) the run meets it.

1.6. Inference

The data support H1, exchangeability of the small-x window with a random base for exact m = 3 coverage, at the tested scope, with about thirteen times finer resolution than the declared delta at C24 (0.5 / 0.038, *derived*). They do not separate E1, E2 and E3 because no arm shows an effect to attribute. 'Constant-only' follows from the conservation ceiling, a theorem independent of the run; the measured content is the narrower exchangeability of this base. The tested base selects abscissae by the integer size of x and never uses the CM structure; a w-orbit base with labels l x lambda^j is constructible (red team, lambda = 2/v mod N for FAM) and was not run. The FAM sign is negative in all three cells; at C24 recorded Holm p is 0.094 (0.187 under the contract text), not a detected deficit, recorded as exploratory.

Why refine and not support. The status a `supported` hypothesis would carry is narrower than the sentence the contract attaches ('constant-only, retire the channel'); the recorded powered set, MDE and Holm values are not the contract's (D1) and no archived analysis applies the contract text yet; the Coordinator's determination that D1 does not fire I9 is a judgment on which the run's standing depends; and the observation is a single unreplicated pass. The data are not read as disagreeing with the null.

1.7. Limitation

Toy sizes 2^14 < p <= 2^25; no transfer to 2^52 or the 53-bit pair; one confirmatory pass, gate numbers seen before the run, planted-gate margins thin at C16 and C20 and one rung only; one powered cell by the code and two by the contract text, no growth arm; FAM is exhaustive in band; the from-scratch rebuild shows determinism, not replication; a coupling growing as N^eps with eps above about 0.13 and below 0.04 percent at 2^24 is unconstrained at 2^52 (red team). Defects of the frozen analysis code, recorded and not corrected: D1 (MDE inflation sqrt(3) x 1.0025 against 'no inflation factor': powered {C24} instead of {C20, C24}, FAM C24 Holm 0.0936 instead of 0.1872, class unchanged) and D2 (I10 against the superseded 0.49/sqrt(N): false positive in 9 of 9). The review plan was written after both verdicts, the blind re-derivation was not blind, two reviewers worked the same quantity, and the reviewed snapshot is a producer-session head. Scan efficiency is validated as an implementation and is immaterial to cost. Distance remaining to matched rho: no measured value; the red team's analytic, unmeasured estimate of index calculus over rho is 88 to 3400 at N = 2^24 across the 0, 0.1 and 1 GAE band with the order holding, quoted as context and not adopted; the small-x base equals the index-calculus baseline (a random base) within 0.04 percent at C24, so no improvement over the baseline was measured.

1.8. Next actions

1. Superseding analysis, zero compute (RC-1): contract-text MDE, powered set, Holm, I10 against 1.4/sqrt(N), per-member z-scores; new script and run id, companion CORR-* record, recorded t8 untouched. This is the one next action for the question.
2. Then a scoped `support` decision on H-CMFB-9b2eed (strength preliminary, wording narrowed per the corrected reading of `decision_impact`), if the new analysis agrees.
3. Commission the design (not approval) of RC-2, a CM-coupled w-orbit base on the same instrument, with the signed convention (RC-3) and a graded dose-response (RC-5) folded in; check overlap with H-CMFB-7d4e1a and H-CMFB-c5e29b first. Low to medium priority; does not displace exponent-targeting work.
4. Not commissioned now: RC-4 (C28 growth cell).
5. Ledger archive of the named paths, validation and independence checks, push, ready-for-review PR naming EV-CMFB-e3d1ec and DEC-20261019-f33f3e.

1.9. Earlier run reports and graphs

Earlier run report (observations only): `experiments/EXP-CMFB-cea1d2/reports/run-20261019T0541Z/` (`report.typ`, `report.pdf`, `fig_run_summary.svg`, `fig_run_summary.png`, `plot.py`).

Canonical graphs checked and left unchanged: `reports/run-20261019T0541Z/fig_run_summary.svg` and `.png` (recorded t8 values; unchanged because the recorded t8 is untouched; the review's view is `fig_review.svg`). `knowledge/frontiers/ecdlp/` holds frontier rows (YAML) and no graph; no row changes because no measured bound or ranking is recorded. No file under `runs/`, `claims/`, `implementation/`, `specification.yaml`, `trial-plan.json` or `reports/run-20261019T0541Z/` was edited. The superseding analysis of action 1 will need its own graph.